Thesenpapier: Dual Spaces, Multilinear Forms, and Quadratic Spaces

Foundational Session 1 — Clifford Seminar

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1 Physical Foundations: Kinematics, Duality, and Measurement

Consider the motion of a single material particle constrained to an affine plane $\mathbb{E}^2$. The set of all kinematically accessible spatial positions forms the configuration manifold M, a smooth two-dimensional manifold diffeomorphic to $\mathbb{R}^2$. The manifold M carries no intrinsic linear structure; points correspond to physical locations rather than vectors, and no distinguished origin exists without an arbitrary reference choice.

At an instantaneous configuration $x \in M$, kinematically admissible generalized velocities $\dot{x}$ do not belong to the manifold M. Instead, they populate the tangent space T_xM, which is an intrinsic two-dimensional real linear space realized algebraically as the space of directional derivations on smooth scalar fields $\mathcal{C}^\infty(M)$. The tangent bundle $TM = \bigsqcup_{x\in M} T_xM$ forms the state space of Lagrangian kinematics, where velocity spaces at distinct base points remain disjoint and cannot be canonically identified without parallel transport.

Generalized forces and canonical momenta p do not reside in T_xM, but in the cotangent space $T_x^*M = \mathrm{Hom}_{\mathbb{R}}(T_xM, \mathbb{R})$, which is the dual linear space of linear functionals on velocities. The canonical evaluation pairing

$$\langle \cdot, \cdot \rangle : T_x^*M \times T_xM \longrightarrow \mathbb{R}, \quad \langle p, v \rangle = p(v), \tag{1}$$

quantifies to twice of the kinetic energy without reference to coordinates or metric tensors.

Physical measurement of kinematic states is fundamentally covectorial. Because a velocity vector cannot be evaluated in isolation, measurement requires coupling with a dual element, such as an apparatus registering differential flux or momentum transfer. In Newtonian mechanics, the identification of a velocity $v \in T_xM$ with a momentum $p \in T_x^*M$ is an auxiliary constitutive mapping supplied by the kinetic quadratic form $T = \frac{1}{2}q(v)$. The associated symmetric bilinear form b obtained through polarization establishes the musical isomorphism

$$b^\flat : T_xM \longrightarrow T_x^*M, \quad v \mapsto b(v, \cdot). \tag{2}$$

A metric is therefore not a geometric primitive of space, but a measurement device converting kinematic vectors into linear functionals.

In relativistic mechanics, planar kinematics is embedded into a real spacetime linear space V. The causal structure of spacetime is established algebraically prior to differential geometry by equipping V with a non-degenerate quadratic form q of signature $(1, 3)$ (or $(1, 2)$ in planar spacetime). The light cone is the algebraic zero-set

$$\mathcal{C} = \{v \in V \mid q(v) = 0\}, \tag{3}$$

partitioning V into timelike ($q(v) > 0$), spacelike ($q(v) < 0$), and null elements. Relativistic momenta reside in the dual space V^*, where the invariant mass-shell condition corresponds to the inverse quadratic form $q^*(p) = m^2c^2$. Causality and measurement are thus shown to be algebraic invariants of quadratic spaces and duality.

2 Foundational Definitions

2.1 Groups, Linear Spaces, and Algebras

Let G be a set equipped with an internal binary operation $* : G \times G \to G$. The pair $(G, *)$ constitutes a group if the operation satisfies the following three conditions:

$\mathcal{G}_1$ Associativity: $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.

$\mathcal{G}_2$ Identity element: there exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.

$\mathcal{G}_3$ Inverse element: for each $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

If $a * b = b * a$ holds for all $a, b \in G$, the group is abelian. An abelian group is written additively as $(V, +)$, with identity element 0 and inverse $-v$.

Let K be a field. A linear space over K is an additive abelian group $(V, +)$ equipped with an external scalar multiplication $\cdot : K \times V \to V$, denoted $(\lambda, v) \mapsto \lambda v$, satisfying the following conditions for all $\lambda, \mu \in K$ and all $u, v \in V$:

1. Distributivity over group addition: $\lambda(u + v) = \lambda u + \lambda v$.
2. Distributivity over field addition: $(\lambda + \mu)v = \lambda v + \mu v$.
3. Compatibility with field multiplication: $\lambda(\mu v) = (\lambda\mu)v$.
4. Unitary action: $1_K v = v$, where 1_K denotes the multiplicative identity of K.

The designation *vector space* is reserved strictly for concrete coordinate realizations such as $\mathbb{R}^n$ or explicit geometric displacement spaces.

Let V be a linear space over K. The structure $(A, +, \cdot, \bullet)$ is a K-algebra if $(A, +, \cdot)$ is a linear space over K equipped with a bilinear binary operation $\bullet : A \times A \to A$, satisfying for all $\lambda \in K$ and $u, v, w \in A$ the compatibility conditions

$$u \bullet (v + w) = u \bullet v + u \bullet w, \quad (u + v) \bullet w = u \bullet w + v \bullet w, \tag{4}$$

and

$$\lambda(u \bullet v) = (\lambda u) \bullet v = u \bullet (\lambda v). \tag{5}$$

The algebra is associative if $(u \bullet v) \bullet w = u \bullet (v \bullet w)$, and unital if there exists an element $1_A \in A$ such that $1_A \bullet u = u \bullet 1_A = u$ for all $u \in A$.

2.2 Dual Spaces and the Canonical Pairing

Let V be a linear space over K. The dual space V^* is the linear space of all linear functionals on V,

$$V^* = \mathrm{Hom}_K(V, K). \tag{6}$$

The evaluation of a functional $\alpha \in V^*$ on an element $v \in V$ is expressed by the bilinear dual pairing

$$\langle \cdot, \cdot \rangle : V^* \times V \to K, \quad \langle \alpha, v \rangle = \alpha(v). \tag{7}$$

The double dual space is $V^{**} = \mathrm{Hom}_K(V^*, K)$. The canonical evaluation mapping $\mathrm{ev} : V \to V^{**}$ defined by

$$\mathrm{ev}_v(\alpha) = \langle \alpha, v \rangle \tag{8}$$

is an injective linear homomorphism. When $\dim V < \infty$, the map ev is an isomorphism. Consequently, elements of V act natively as linear functionals on V^* without auxiliary structure.

2.3 The Free Tensor Algebra

The tensor algebra of a linear space V over K is the graded associative K-algebra

$$T(V) = \bigoplus_{r=0}^{\infty} T^r(V), \tag{9}$$

where $T^0(V) = K$, $T^1(V) = V$, and $T^r(V) = \bigotimes^r V$ for $r \geq 2$. Multiplication is governed by the associative tensor product $\otimes$. The structure $T(V)$ satisfies a universal property: every linear map from V into an associative unital K-algebra A extends uniquely to a unital algebra homomorphism from $T(V)$ into A. The tensor algebra imposes no algebraic relations between distinct elements of V.

2.4 Quadratic Forms and Polarization

Let V be a real linear space. A map $q : V \to \mathbb{R}$ is a quadratic form if it satisfies $q(\alpha v) = \alpha^2 q(v)$ for all $\alpha \in \mathbb{R}$ and $v \in V$, and the associated symmetric bilinear form $b : V \times V \to \mathbb{R}$ defined by the polarization identity

$$b(u, v) = \frac{1}{2}\big(q(u + v) - q(u) - q(v)\big) \tag{10}$$

is linear in each argument. The radical of the quadratic space (V, q) is the subspace

$$\operatorname{rad}(V) = \{u \in V \mid b(u, v) = 0 \text{ for all } v \in V\}. \tag{11}$$

The space (V, q) is non-degenerate if $\operatorname{rad}(V) = \{0\}$.

A symmetric bilinear form induces a linear mapping $b^\flat : V \to V^*$ given by $b^\flat(u) = b(u, \cdot)$. Non-degeneracy of q is equivalent to $b^\flat$ being an isomorphism, whose inverse is denoted $b^\sharp : V^* \to V$.

By Sylvester's law of inertia, any finite-dimensional real quadratic space decomposes into an orthogonal direct sum $V = V_+ \oplus V_- \oplus V_0$ such that $q|_{V_+}$ is positive definite, $q|_{V_-}$ is negative definite, and $q|_{V_0} = 0$. In the non-degenerate case, $\dim V_0 = 0$, and the signature is denoted (p, m), where $p = \dim V_+$ and $m = \dim V_-$. The resulting pseudo-Euclidean vector space is denoted $\mathbb{R}^{p,m}$.

3 Derivation Agenda for the Blackboard Exposition

The 40-minute Referat follows four sequential stages:

1. Axiomatic synthesis of linear spaces from additive abelian groups and external field actions, followed by the coordinate-free definition of V^* and the canonical isomorphism $V \cong V^{**}$.

2. Construction of the free tensor algebra $T(V)$ as the maximal multilinear structure, demonstrating its infinite-dimensional, relation-free nature.

3. Introduction of the quadratic form q, derivation of the symmetric bilinear form via the polarization identity, and the construction of the musical isomorphisms $b^\flat$ and $b^\sharp$.

4. Orthogonal decomposition of non-degenerate quadratic spaces, proof of signature invariance, and classification of isotropic rays $q(v) = 0$.

4 Theses for Plenary Discussion

Thesis I The standard textbook practice of identifying velocities with canonical momenta via Cartesian index identification ($p_i = m\dot{x}_i$) obscures the duality between V and V^*. This obscures the geometric foundation of Hamiltonian mechanics and creates conceptual barriers when transitioning to curved manifolds or relativistic field theories.

Thesis II The free tensor algebra $T(V)$ contains excess degrees of freedom that lack geometric meaning. A physical geometry cannot emerge from multilinear algebra until an algebraic ideal generated by measurement—specifically the quadratic form $q(v)$—is factored out.

Thesis III A metric is not an intrinsic property of a linear space, but an auxiliary constitutive mapping $b^\flat : V \to V^*$ that converts displacements into linear measurement functionals.

Thesis IV Causal structure in relativistic physics is an algebraic consequence of the signature $(1, 3)$ on a four-dimensional linear space, independent of differential geometry or pseudo-Riemannian metric tensors.

5 Preparatory Exercises

1. Consider the set $\mathcal{U}$ of coherent physical units generated by the base units of the International System of Units (SI). Let group addition on $\mathcal{U}$ be defined by physical unit multiplication (for example, $\mathrm{m} \cdot \mathrm{s} = \mathrm{m\,s}$ and $\mathrm{m} \cdot \mathrm{m} = \mathrm{m}^2$), and define external scalar multiplication by $\mathbb{Q}$ as unit exponentiation, $(\lambda, u) \mapsto u^\lambda$. Show that $(\mathcal{U}, \cdot)$ satisfies the axioms of an additive abelian group, with the dimensionless unit 1 serving as identity, and verify that this structure satisfies all axioms of a linear space over $\mathbb{Q}$.

2. Let the ordered set of seven base units $\mathcal{B} = \{\mathrm{m}, \mathrm{kg}, \mathrm{s}, \mathrm{A}, \mathrm{K}, \mathrm{mol}, \mathrm{cd}\}$ form a basis for the SI linear space over $\mathbb{Q}$. Determine the explicit coordinate tuples in $\mathbb{Q}^7$ corresponding to:

   (a) The Newton (N), unit of mechanical force.

   (b) The Farad (F), unit of electrical capacitance.

   (c) The Lumen (lm), unit of luminous flux.

3. Let V be a finite-dimensional linear space over K, and let $\mathcal{E} = \{e_i\}_{i=1}^n$ be a basis for V. The dual basis $\mathcal{E}^* = \{\omega^j\}_{j=1}^n \subset V^*$ is defined by $\langle \omega^j, e_i \rangle = \delta_i^j$. Show that the mapping $e_i \mapsto \omega^i$ depends explicitly on the choice of basis, confirming that no canonical isomorphism between V and V^* exists. In contrast, prove that the evaluation map $\mathrm{ev} : V \to V^{**}$ defined by $\mathrm{ev}_v(\alpha) = \langle \alpha, v \rangle$ is completely basis-independent and canonical.

4. In classical mechanics, consider a free particle on $\mathbb{R}^3$ described by the Lagrangian $L(v) = \frac{1}{2} m b(v, v)$, where b is a symmetric, positive-definite bilinear form. Show that the definition of canonical momentum $p = \partial_v L$ is identical to the action of the musical isomorphism $b^\flat : V \to V^*$. Compute $b^\sharp : V^* \to V$ and show that the classical Hamiltonian $H(p) = \langle p, v \rangle - L(v)$ corresponds to the inverse quadratic form on the dual space.

5. Consider the real linear space $V = \mathbb{R}^4$ equipped with the Minkowski quadratic form $q(x) = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$.

   (a) Apply the polarization identity to determine the explicit symmetric bilinear form $b(x, y)$.

   (b) Show that $\mathrm{rad}(V) = \{0\}$, establishing non-degeneracy.

   (c) A linear subspace $W \subset V$ is totally isotropic if $q|_W = 0$. Prove that the maximum dimension of any totally isotropic subspace in $\mathbb{R}^{1,3}$ is 1, and explain why physical light rays cannot form a two-dimensional planar subspace.

6 Preparatory Literature

Participants are expected to study the following references prior to the session:

- Abstract groups and linear spaces: Fraleigh [1, Chaps. 1.1, 1.2, and 7.33]; Axler [2, Chaps. 1 and 2].
- Dual spaces, multilinear forms, and quadratic spaces: Vaz and da Rocha [3, Chap. 1].

References

[1] John B. Fraleigh and Victor J. Katz. *A First Course in Abstract Algebra*. 8th ed. World Student Series. Hoboken, New Jersey: Pearson, 2021. 424 pp. ISBN: 978-0-13-575816-8.

[2] Sheldon Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Cham: Springer International Publishing, 2015. ISBN: 978-3-319-11079-0. DOI: 10.1007/978-3-319-11080-6. URL: https://link.springer.com/10.1007/978-3-319-11080-6.

[3] Jayme Vaz and Roldão da Rocha. *Introduction to Clifford Algebras and Spinors*. Oxford: Oxford university press, 2016. 128 pp. ISBN: 978-0-19-878292-6.